BESANT TECHNOLOGIES
(BTM LAYOUT)

DATA ANALYSIS PROJECT

BANKING CUSTOMER TRANSACTION AND CREDIT RISK ANALYSIS

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UNDER THE GUDANCE OF
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1. Introduction

The banking industry generates vast amounts of customer data related to demographics, transactions, credit usage, and assets. Analyzing this data helps financial institutions assess customer risk, improve service offerings, and enhance profitability. This project analyzes a structured banking dataset containing customer demographic details, income, loans, deposits, credit card usage, properties owned, and risk weighting.

The objective is to understand customer demographics, financial behavior, credit usage, and risk patterns through structured data analysis and visualization. The study demonstrates how data-driven insights can support better decision-making in the banking sector, particularly in customer segmentation, credit risk management, and strategic planning.

2. Objectives of the Analysis

I. Objectives.

- **To study customer details** such as age, occupation, nationality, and loyalty level to understand the bank's customer base.
- **To analyse customer financial activities** including income, deposits, loans, and credit card usage.
- **To identify the main factors that affect credit risk** by examining the relationship between risk level, loans, deposits, and credit card balances.
- **To explore patterns and trends in the data** using exploratory data analysis techniques.
- **To understand customer loyalty behaviour** by comparing loyalty levels with income, asset ownership, and banking product usage.
- **To analyse customer joining trends over time** and observe changes in financial behaviour across different years.
- **To provide useful insights** that help banks improve customer segmentation, manage risk, and make better business decisions.

II. Problem Statement

1. **What is the relationship between customer income and bank loan amounts?**

   The correlation analysis shows a **positive correlation (0.33)** between **Estimated Income** and **Bank Loans**.
   - This indicates that as **customer income increases, loan amounts also tend to increase**.
   - However, the correlation is **moderate**, not strong, meaning income is **not the only factor** influencing loan borrowing.

   .

	Estimated Income	Bank Loans
Estimated Income	1.000000	0.329926
Bank Loans	0.329926	1.000000

Interpretation: Higher-income customers are generally more eligible and confident to take loans, but loan decisions are also affected by factors such as **credit history, risk weighting, assets owned, and banking relationships**.

2. **Which factors contribute most to higher credit risk among customers?**

As **risk weighting increases from 1 to 5**, both **average bank loans** and **credit card balances** show a **consistent upward trend**. Customers in the **highest risk category (Risk Weighting 5)** have the **largest loan amounts and highest credit card balances**, indicating heavier borrowing behaviour. Lower risk categories maintain significantly lower credit exposure.

```
df.groupby('Risk Weighting')[['Bank Loans', 'Credit Card Balance']].mean()
```

Risk Weighting	Bank Loans	Credit Card Balance
1	334192.148553	1882.421459
2	575784.755401	3081.658723
3	786634.906804	4110.491435
4	848689.886770	4504.116584
5	975216.615625	5299.862063

Interpretation: The results indicate a **strong positive relationship between risk level and credit exposure**. Customers with higher risk weighting tend to rely more on loans and credit cards, increasing the bank's potential default risk. This insight highlights the importance of **closer monitoring, stricter credit policies, and proactive risk management** for high-risk customer segments.

3. **How does customer loyalty classification vary with income levels?**

- Silver **and Jade customers** have the **highest average income**, indicating that higher income does not always translate into higher loyalty tiers.
- **Gold customers** show a moderately high income level.
- **Platinum customers**, despite being a premium loyalty segment, have slightly lower average income compared to other groups.

```
df.groupby('Loyalty Classification')['Estimated Income'].mean()
```

```
Loyalty Classification
Gold        167876.148718
Jade        172418.944673
Platinum    163760.178202
Silver      175105.559244
Name: Estimated Income, dtype: float64
```

Interpretation: The results suggest that **loyalty classification is influenced by factors beyond income**, such as transaction frequency, product usage, and relationship duration. Therefore, banks should not rely only on income while assigning loyalty tiers but should consider **overall customer engagement and value**.

4. **How does asset ownership affect bank deposits?**

- Customers owning **2 properties** have the **highest average bank deposits**, indicating stronger financial capacity.
- Customers with **0 or 1 property** also maintain high deposits, showing that property ownership alone does not strictly determine savings behavior.
- Customers owning **3 properties** show slightly lower average deposits, possibly due to higher financial commitments such as loans or maintenance costs.

```
df.groupby('Properties Owned')['Bank Deposits'].mean()
```

```
Properties Owned
0    686327.653518
1    656519.965915
2    698634.218468
3    644907.418787
Name: Bank Deposits, dtype: float64
```

Interpretation: Overall, **property ownership has a moderate influence on bank deposits**, and higher property count does not always result in higher savings. This insight suggests that banks should consider **multiple financial factors**, not just asset ownership, when evaluating customer wealth and investment potential.

5. **Can high-risk and high-value customers be identified using financial data?**

High-value customers were identified as those whose income and bank deposits fall in the top 25%, indicating strong financial stability and high contribution to the bank.
High-risk customers were identified as those with high risk weighting (top 25%) and loan amounts exceeding deposits, indicating higher credit exposure.

```
high_value = df[
    (df['Estimated Income'] > df['Estimated Income'].quantile(0.75)) &
    (df['Bank Deposits'] > df['Bank Deposits'].quantile(0.75))
]

high_risk = df[
    (df['Risk Weighting'] > df['Risk Weighting'].quantile(0.75)) &
    (df['Bank Loans'] > df['Bank Deposits'])
]
print("High Value Customers:", high_value.shape[0])
print("High Risk Customers:", high_risk.shape[0])
```

```
High Value Customers: 294
High Risk Customers: 243
```

Interpretation:

- 294 high-value customers represent profitable and financially stable clients suitable for premium services.
- 243 high-risk customers indicate potential credit risk and require closer monitoring and risk control measures.

III. Outcome of the Analysis:

1. The analysis effectively segmented customers into high-value and high-risk groups using financial and risk indicators.
2. Estimated income shows a moderate positive relationship with bank loan amounts.
3. Higher risk-weighted customers carry significantly greater loan and credit card balances.
4. Property ownership and loyalty levels alone do not directly determine customer wealth.
5. The insights obtained support improved customer targeting, risk control, and strategic banking decisions.

3. Data Collection

I. Gathering & Transfer

- The dataset (*Banking.csv*) was first imported into **MySQL** to ensure structured, scalable storage and to simulate a real-world **ETL (Extract, Transform, Load)** workflow.
- A relational table was created with key fields such as Client ID, Name, Age, Location ID, Joined Bank, Banking Contact, Nationality, Occupation, Fee Structure, Loyalty Classification, Estimated Income, Superannuation Savings, Amount of Credit Cards, Credit Card Balance, Bank Loans, Bank Deposits, Checking Accounts, Saving Accounts, Foreign Currency Account, Business Lending, Properties Owned, Risk Weighting, BRId, GenderId, IAId using appropriate data types for accuracy and performance.
- Superannuation Savings, using appropriate data types for accuracy and performance.
- The stored data was then connected to **Jupiter Notebook** using **SQLAlchemy and MySQL** and loaded into **pandas Data Frames** via read_sql () for further cleaning, analysis, and visualization.

```
df = pd.read_csv("D:\project\Banking\Banking.csv")
df.head(5)
```

	Client ID	Name	Age	Location ID	Joined Bank	Banking Contact	Nationality	Occupation	Fee Structure	Loyalty Classification	...	Bank Deposits	Checking Accounts	Saving Accounts	Foreign Currency Account
0	IND81288	Raymond Mills	24	34324	06-05-2019	Anthony Torres	American	Safety Technician IV	High	Jade	...	1485828.64	603617.88	607332.46	12249.96
1	IND65833	Julia Spencer	23	42205	10-12-2001	Jonathan Hawkins	African	Software Consultant	High	Jade	...	641482.79	229521.37	344635.16	61162.31
2	IND47499	Stephen Murray	27	7314	25-01-2010	Anthony Berry	European	Help Desk Operator	High	Gold	...	1033401.59	652674.69	203054.35	79071.78
3	IND72498	Virginia Garza	40	34594	28-03-2019	Steve Diaz	American	Geologist II	Mid	Silver	...	1048157.49	1048157.49	234685.02	57513.65
4	IND60181	Melissa Sanders	46	41269	20-07-2012	Shawn Long	American	Assistant Professor	Mid	Platinum	...	487782.53	446644.25	128351.45	30012.14

5 rows × 25 columns

II. Integrity of the Data

- The dataset contains **3,000 rows and 25 columns**, ensuring sufficient data for meaningful analysis.
- All fields have **zero null values**, indicating complete and reliable records.
- Data types are properly defined (numeric fields for Age, Estimated Income, Bank Loans, Risk Weighting; text fields for categorical attributes).
- The clean and structured dataset supports accurate analysis and valid insights.

4. Data Inspection/ Initial Analysis

I. Details of Data

- **Dataset Overview:** The dataset consists of **3,000 customer records** with **25 columns**, providing a comprehensive view of customer banking behavior.
- **Column Review:** The dataset includes both categorical and numeric variables, such as customer identity details, income, deposits, loans, credit card usage, property ownership, and risk weighting**.**
- **Missing Values:** There are no null values across any columns, indicating complete records.
- **Data Types:** Numeric columns are correctly set as int64 or float64, and categorical columns are object type, suitable for analysis.

II. Null Value analysis

- There are no null and duplicated values in the Banking Customer Transaction dataset.

```
df.isna().sum()
```

```
Client ID                   0
Name                        0
Age                         0
Location ID                 0
Joined Bank                 0
Banking Contact             0
Nationality                 0
Occupation                  0
Fee Structure               0
Loyalty Classification      0
Estimated Income            0
Superannuation Savings      0
Amount of Credit Cards      0
Credit Card Balance         0
Bank Loans                  0
Bank Deposits               0
Checking Accounts           0
Saving Accounts             0
Foreign Currency Account    0
Business Lending            0
Properties Owned            0
Risk Weighting              0
BRId                        0
GenderId                    0
IAId                        0
dtype: int64
```

5. Data Cleaning and Transformation

I. Data Preprocessing

- Checked for missing values and confirmed all columns are complete (0 nulls).
- Data types were verified to ensure numeric and categorical fields were correctly classified.
- Date-related fields, such as the *Joined Bank* column, were converted into a proper date format to enable time-based analysis
- Standardized categorical values for consistency.
- Duplicate records were also examined to maintain data integrity.
- Converted data into pandas Data Frame for efficient analysis and visualization in Jupiter Notebook.
- Ensured dataset is ready for statistical analysis, visualization, and machine learning modelling.

```
df.duplicated().sum()
```

```
np.int64(0)
```

II. Date Column Handling

- Banking Customer Transaction Data contains a date-related column, **Joined Bank**, which represents the date when a customer joined the bank.
- This column was converted from its original string format into a proper **date/datetime format** during preprocessing.

```
df['Joined Bank'].head()
```

```
0    06-05-2019
1    10-12-2001
2    25-01-2010
3    28-03-2019
4    20-07-2012
Name: Joined Bank, dtype: object
```

```
df['Joined Bank'] = pd.to_datetime(df['Joined Bank'], errors='coerce')
```

```
df[['Joined Bank']].dtypes
```

```
Joined Bank    datetime64[ns]
dtype: object
```

6. Exploratory Data Analysis (EDA)

I. Statistical Overview

- Descriptive statistics were computed for age, income, loans, deposits, and credit card balances. This provided an understanding of central tendencies and dispersion.

```
df.describe()
```

	Age	Location ID	Estimated Income	Superannuation Savings	Amount of Credit Cards	Credit Card Balance
count	3000.000000	3000.000000	3000.000000	3000.000000	3000.000000	3000.000000
mean	51.039667	21563.323000	171305.034263	25531.599673	1.463667	3176.206943
std	19.854760	12462.273017	111935.808209	16259.950770	0.676387	2497.094709
min	17.000000	12.000000	15919.480000	1482.030000	1.000000	1.170000
25%	34.000000	10803.500000	82906.595000	12513.775000	1.000000	1236.630000
50%	51.000000	21129.500000	142313.480000	22357.355000	1.000000	2560.805000
75%	69.000000	32054.500000	242290.305000	35464.740000	2.000000	4522.632500
max	85.000000	43369.000000	522330.260000	75963.900000	3.000000	13991.990000

II. Feature Analysis

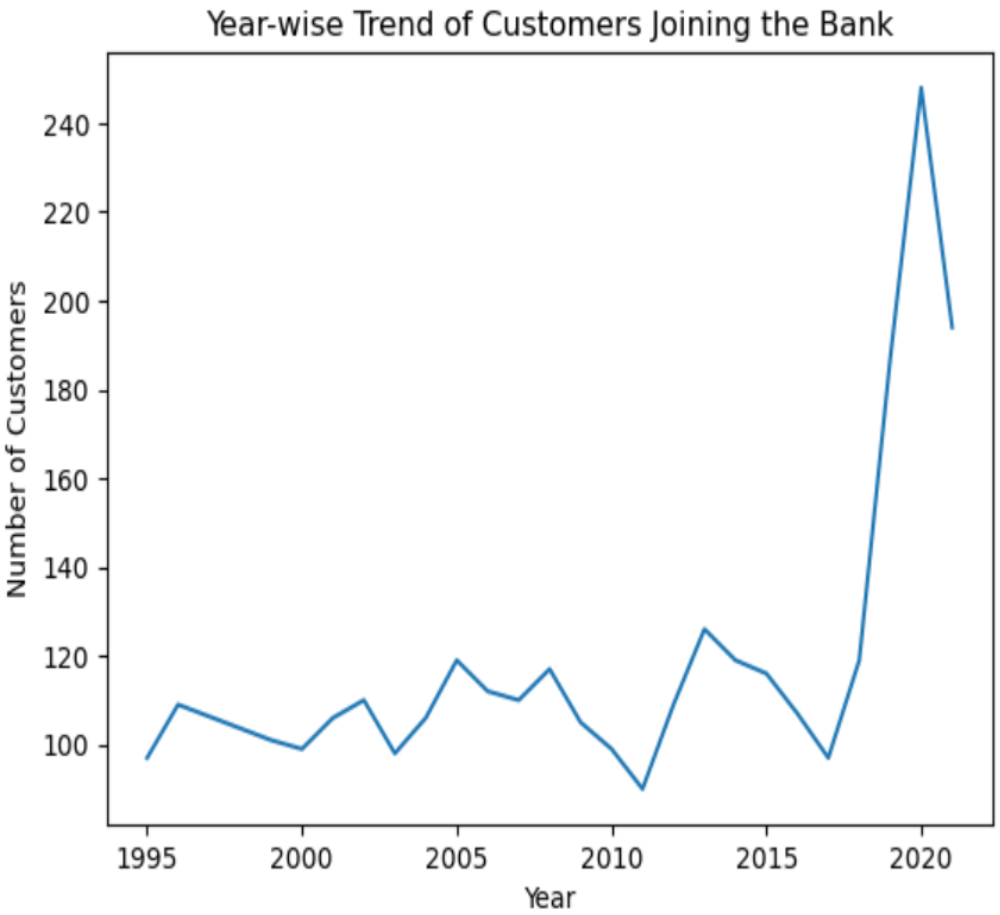
- Histograms & boxplots for all features (matplotlib & seaborn).
- Outlier and spread shown in boxplots.

III. Relationship & Correlation Analysis

- **Bank Loans vs Bank Deposits:** Customers with high loans & low deposits cluster separately
  Financially healthier customers maintain balanced deposits
- **Credit Card Balance vs Risk Weighting:** High-risk customers have significantly higher credit card balances. Indicates higher credit utilization
- **Income vs Credit Risk :** High-risk customers have lower median income
  Low-risk customers show higher income stability
- High **credit card balance + low deposits** increases risk
- Customers with **multiple properties** show lower risk
- **Loyalty classification** is a strong indicator of risk behaviour
- Higher income alone does not guarantee low risk
- New customers show more financial volatility

7. Visualization

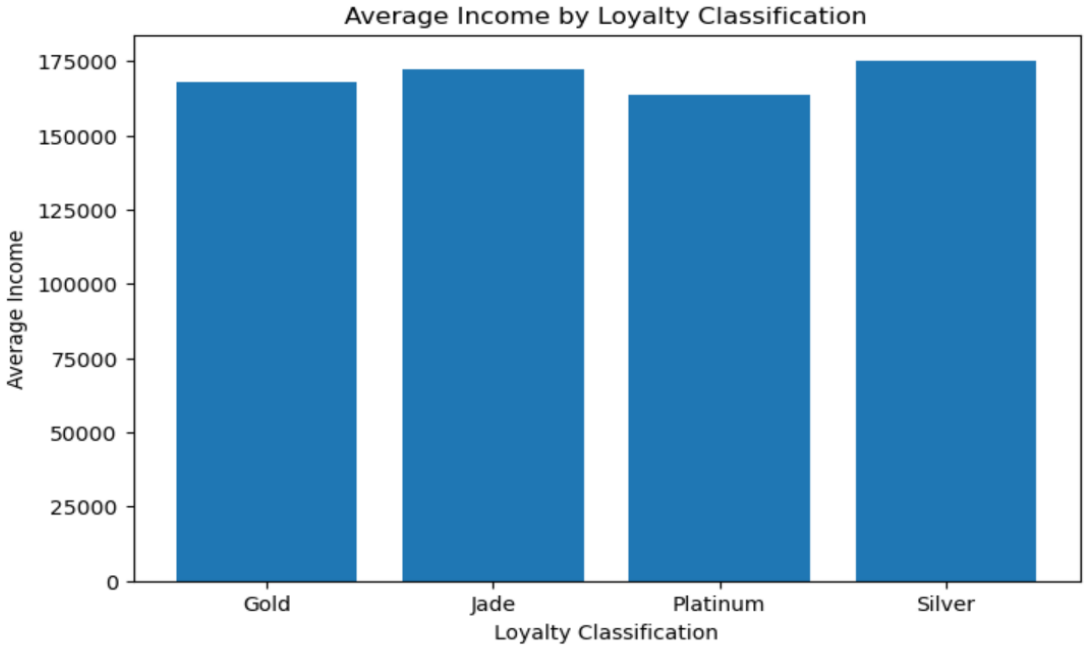
I. **How has customer onboarding changed over time? (Year-wise trend of customers joining the bank) (Line Plot).**



Insights:

- Customer on boarding shows **fluctuating trends** across years
- A **sharp increase around 2020** indicates aggressive customer acquisition or digital banking adoption
- Slight decline after peak suggests market stabilization
- Bank growth is **not linear**, indicating external and internal influencing factors

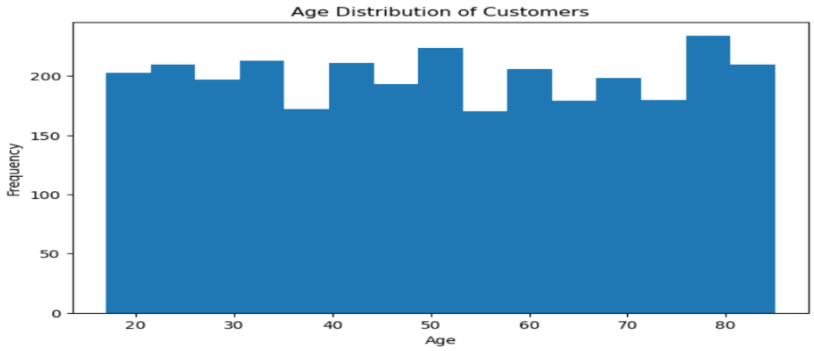
II. How does customer loyalty classification relate to average income levels? (Bar Chart)



Insights

- **Higher loyalty does** not strictly increase with income.
- **Silver customers** show the highest average income
- **Platinum customers** have the lowest average income
- **Gold and Jade** lie in between with similar income levels

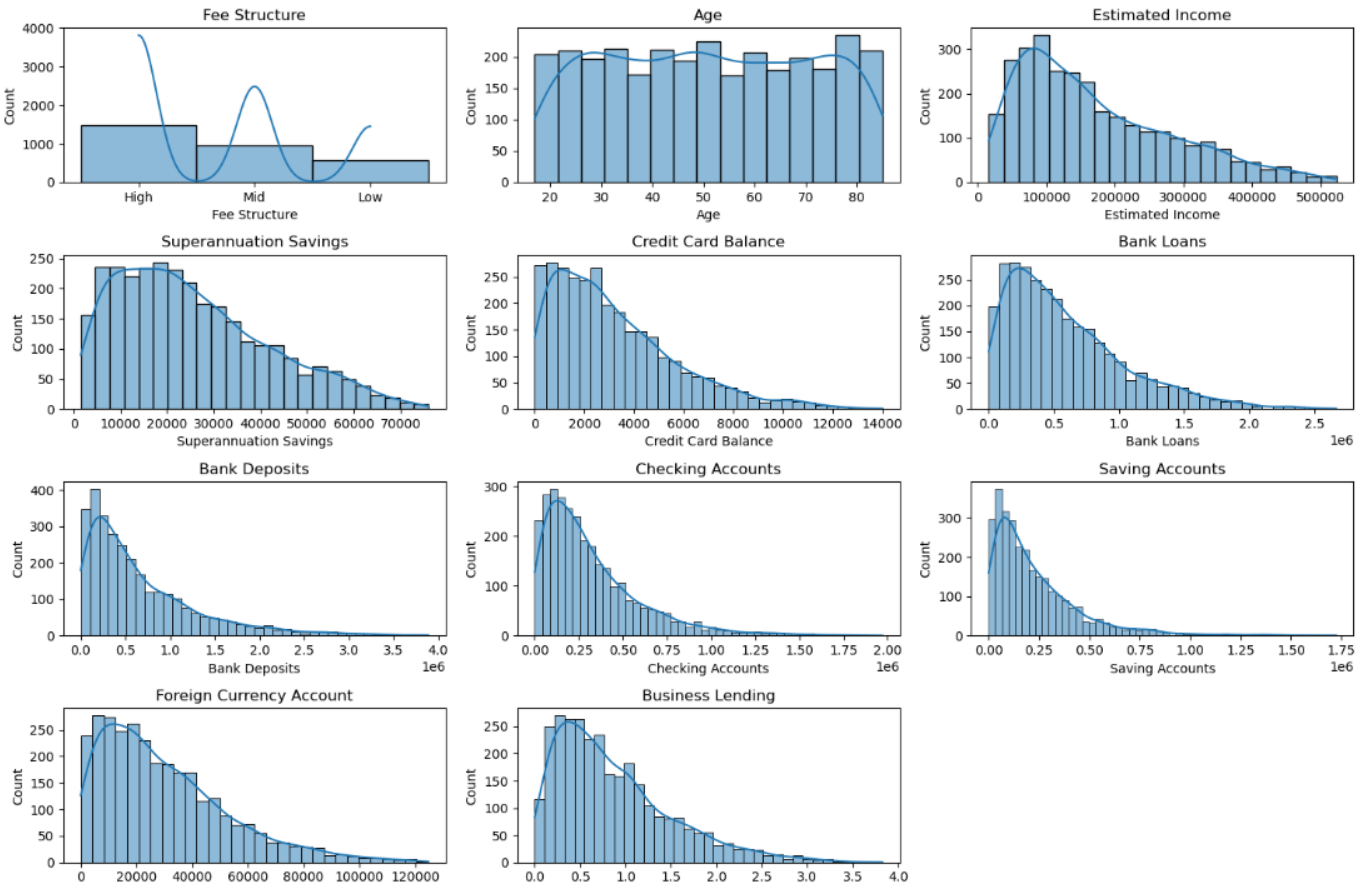
III. What is the age distribution of the bank's customers, and which age groups are most represented in the dataset?



Insights:

- The age distribution shows a well-balanced customer base ranging from young adults to senior citizens, indicating broad market coverage and minimal age concentration bias**.**
- The customer base spans a **diverse age range**, supporting inclusive banking analysis.
- The balanced distribution ensures **robust exploratory and statistical analysis**.
- Different age segments can be targeted with **age-specific financial products**.

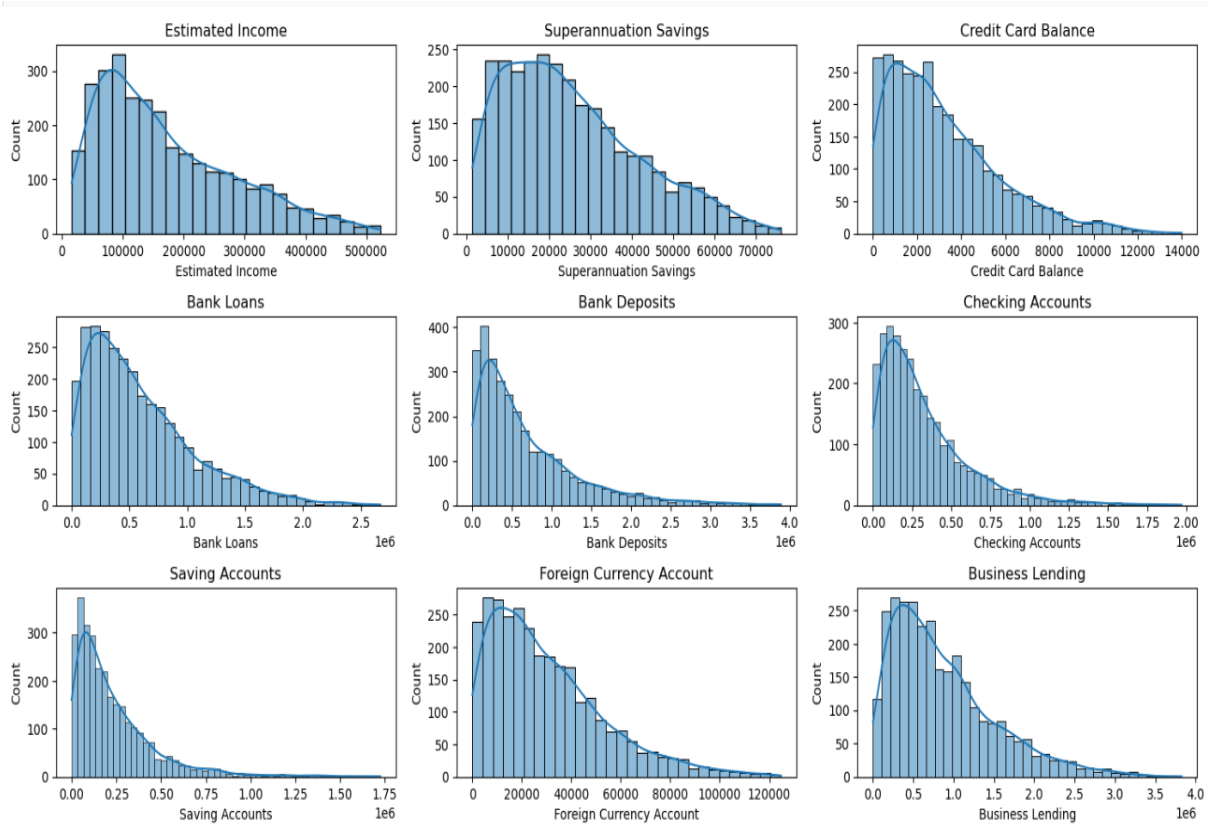
IV. "How are key customer demographic, banking relationship, and financial features distributed across the customer base?" (Histogram with KDE)



Insights:

- **Fee Structure:** Majority of customers fall under mid fee structure. High and Low fee customers are fewer.
- **Age:** Age is fairly uniformly distributed. Customers range broadly from young adults to seniors
- **Estimated Income:** Right-skewed distribution. Most customers earn in lower-to-mid income range Few very high-income outliers
- **Superannuation Savings:** Strong right skew. Majority have modest retirement savings. Long tail of high savers.
- **Credit Card Balance:** Highly positively skewed. Most customers maintain low balances. Small group carries very high balances.
- **Bank Loans:** Right-skewed with extreme values. Majority have smaller loan amounts. Few customers hold very large loans.
- **Bank Deposits:** Strong skew with heavy tail. Majority maintain lower deposit balances.
- **Checking Accounts:** Most customers maintain low checking balances. Sharp drop as balance increases.
- **Saving Accounts:** Right-skewed similar to deposits. Majority save smaller amounts.
- **Foreign Currency Account:** Most customers have low foreign currency holdings. Few customers actively engage in forex.
- **Business Lending:** Heavily skewed. Majority have minimal business lending. Few large exposure accounts.

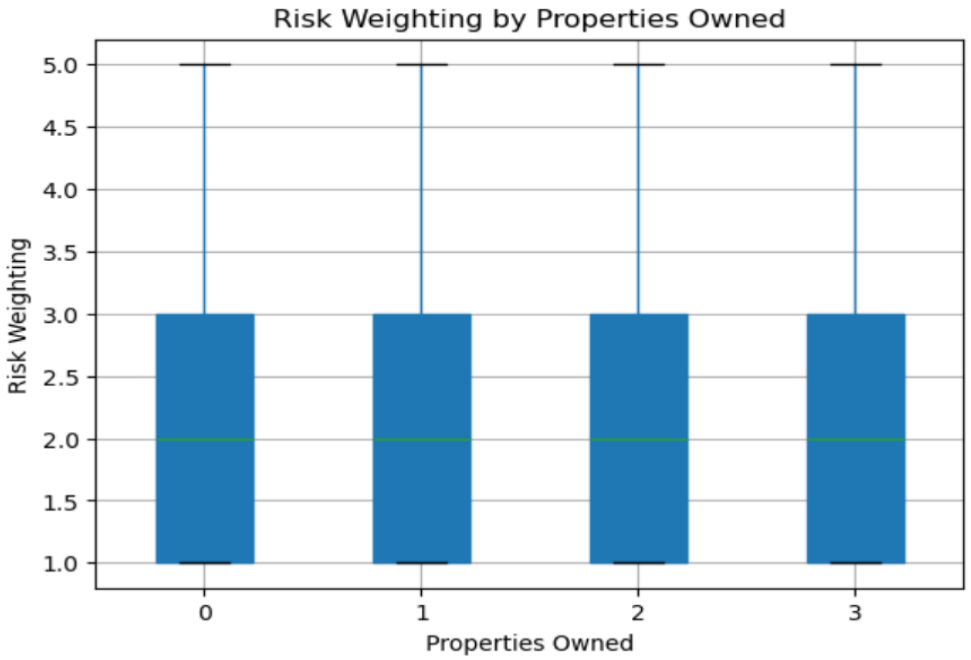
V. How are customers' financial product balances distributed across different banking features?



Insights:

- **Estimated Income:** Strong **right-skewed distribution.** Income inequality exists within customer base. High-income customers form a small elite segment.
- **Superannuation Savings: Right-skewed.** Most customers have moderate retirement savings.
- **Credit Card Balance: Highly right-skewed**. Most customers maintain low balances. Indicates **credit utilization risk.**
- **Bank Loans: Long-tailed distribution**. Few customers hold extremely large loans
- **Bank Deposits: Heavy right skew.** Majority keep small deposits.
- **Checking Accounts:** Sharp drop after low balances.
- **Saving Accounts:** Similar to deposits but lower values.
- **Foreign Currency Account: Right-skewed with long tail.** Used by niche customers (business, overseas exposure)
- **Business Lending: Strong skewness**. Few customers hold large business loans

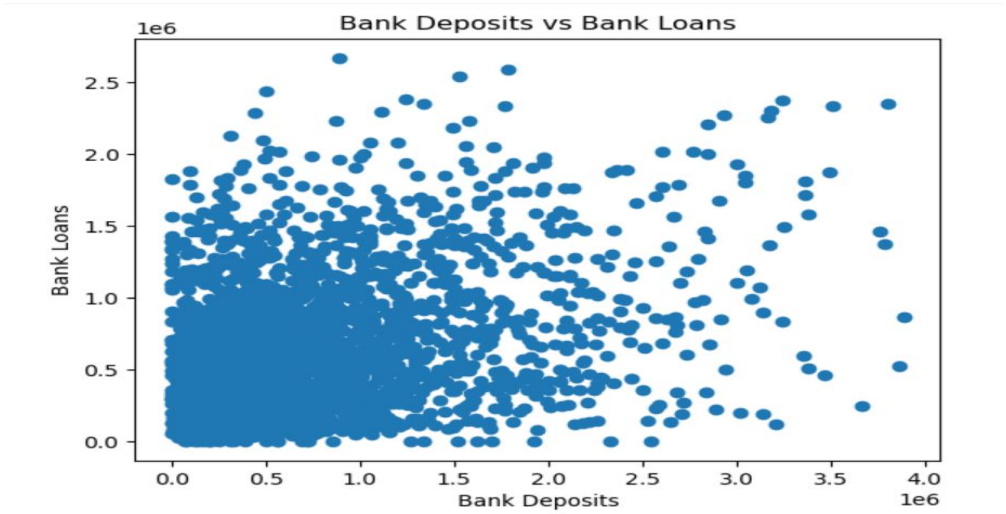
VI. Does owning more properties reduce customer risk? (Box Plot)



Insights:

- Risk weighting slightly decreases for customers owning properties
- Property ownership contributes to better risk profile but is not the only factor

VII. What is the relationship between bank deposits and bank loans?



Insight:

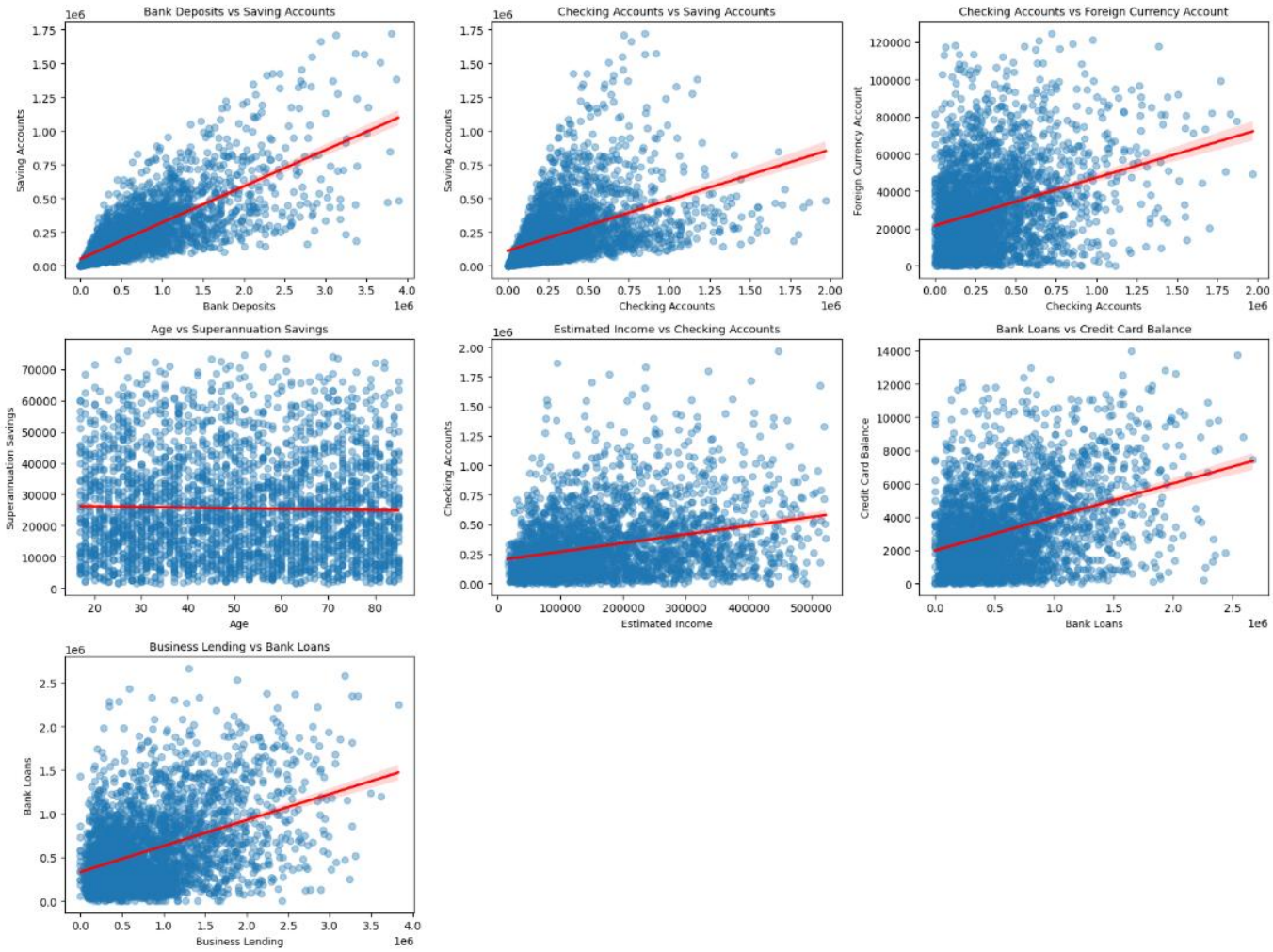
- No strong linear relationship
- Customers with low deposits can still have high loan amounts
- Indicates diverse credit behaviour

VIII. Correlation Heat map (Numerical Features)

Insights

- The correlation matrix highlights strong relationships among deposit and account balance variables, while income moderately influences savings and lending behavior, providing valuable insights into customer financial engagement and credit exposure.
- **Deposits, checking, and savings accounts** form a tightly connected cluster representing customer liquidity.
- **Loans, credit card balances, and business lending** indicate borrowing intensity and credit exposure.
- **Income** acts as a central driver influencing savings, lending, and advanced banking services.
- The absence of extreme correlations suggests minimal multi colinearity risk (except deposit-related features).

IX. How do key banking variables relate to each other, and what kind of linear relationships exist between deposits, savings, income, loans, credit usage, and account balances? (Scatter Plot)



Insights

- The scatter plots reveal strong positive relationships among savings and deposit-related variables, moderate income-driven liquidity patterns, and clustered credit usage behaviour, while age shows minimal influence on retirement saving.

- **Income influences liquidity**, but not perfectly—other factors also play a role.

- **Savings and deposits** show the strongest relationships, reflecting consistent saving behavior.

- **Credit products (loans, credit cards, business lending)** are closely linked, indicating clustered borrowing behaviour.

- **Age is not a strong determinant** of superannuation savings in this dataset.

8. Insights Generation

- The customer base mainly consists of **working-age individuals**, indicating strong potential for long-term banking relationships.
- Customer onboarding has **grown over time**, showing effective customer acquisition with minor fluctuations.
- **Loyalty classification does not strictly depend on income**, suggesting engagement and tenure also influence loyalty tiers.
- Customers holding **multiple credit cards show higher credit balances**, increasing overall credit risk.
- There is **no strong correlation between deposits and loans**, indicating loans are granted based on multiple financial factors.
- Most customers fall under **low to medium risk categories**, reflecting a stable banking portfolio.
- Customers owning **properties generally show lower risk**, indicating better financial stability.
- Higher loyalty customers maintain **greater deposits and savings**, making them valuable targets for premium services.

9. Tools & Technologies Used

- **Python:** Main language for data analysis and modelling.
    - **Pandas, NumPy:** Data handling, transformation, statistics.
    - **Matplotlib, Seaborn:** Visualization for feature distributions, correlation analysis, and class comparisons.
- **Jupiter Notebook:** Interactive development environment for coding, documentation, and result sharing.
- **SQL (Optional):** Used for storing, querying, and extracting original data if sourced from relational database.
- **Gemini AI:** Used for project formats and confusion clarification.

10. Conclusion

This analysis provides a comprehensive understanding of customer demographics, financial behavior, risk exposure, and loyalty patterns. The findings indicate that while the bank maintains a stable risk portfolio, opportunities exist to enhance customer segmentation, credit risk monitoring, and loyalty program effectiveness. Leveraging these insights can help the bank improve profitability, reduce risk, and strengthen long-term customer relationships.